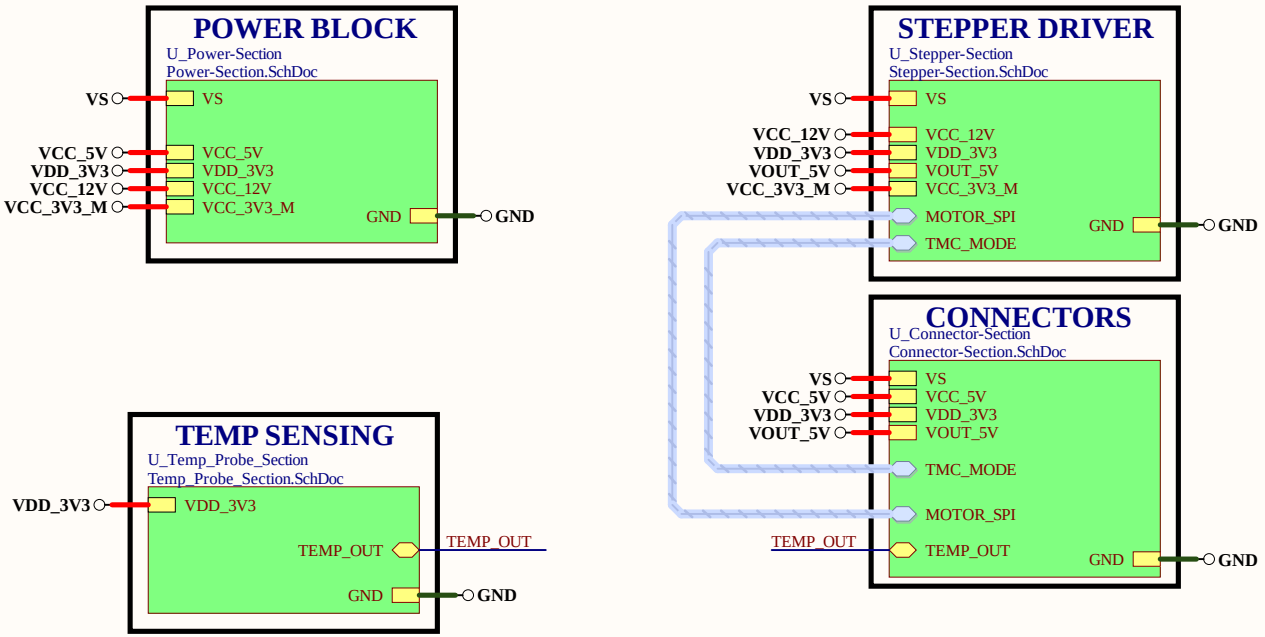
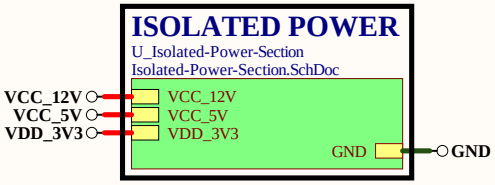
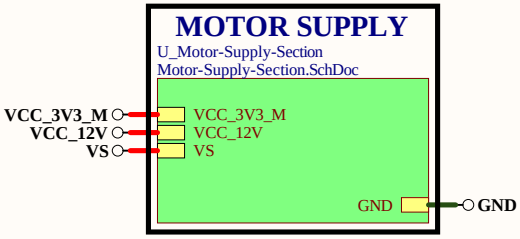
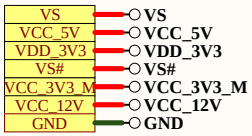
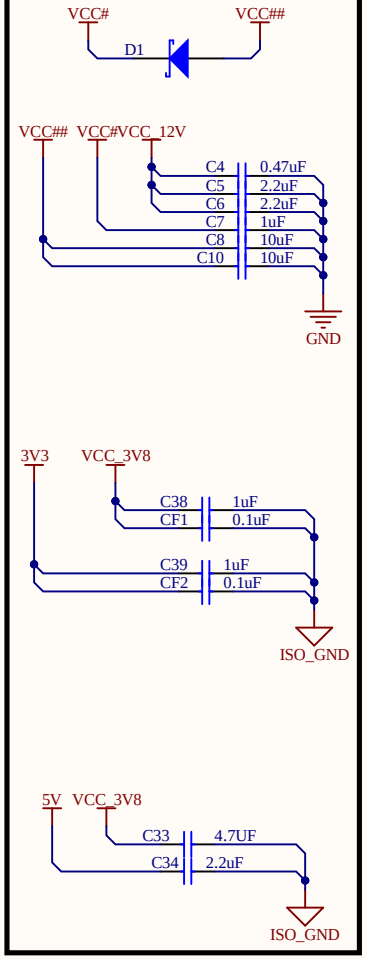

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
MOTOR_SPI
TMC_MODE
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
TEMP_OUT


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
VCC_12V
VCC_5V
VDD_3V3
GND
GND

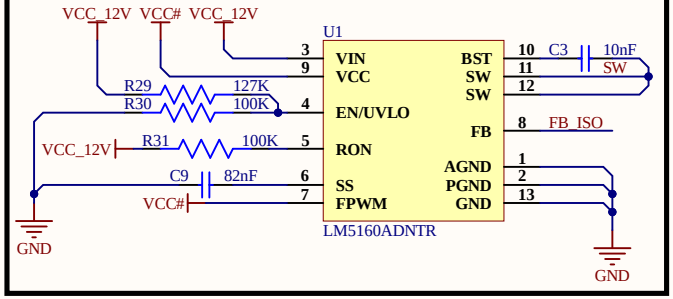

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VCC_3V3_M
VCC_12V
VS
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

DECOUPLING CAPS

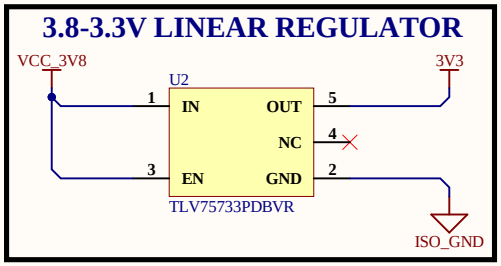


12V-3.8V ISOLATED BUCK CONVERTER

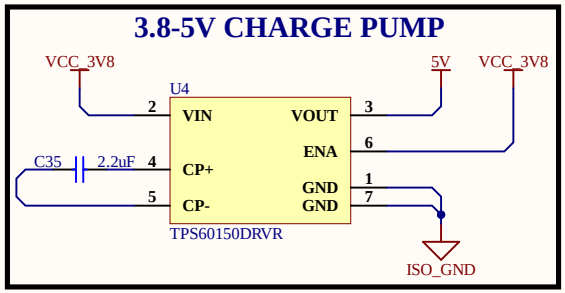


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

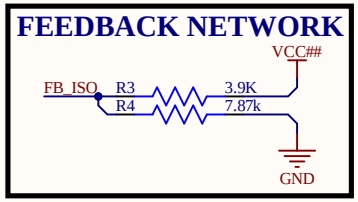
3.8-3.3V LINEAR REGULATOR



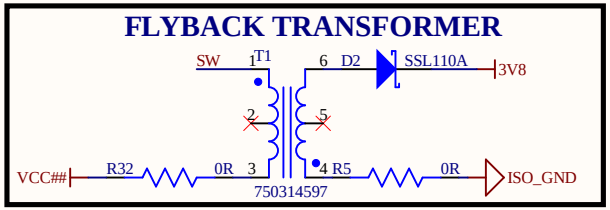
3.8-5V CHARGE PUMP



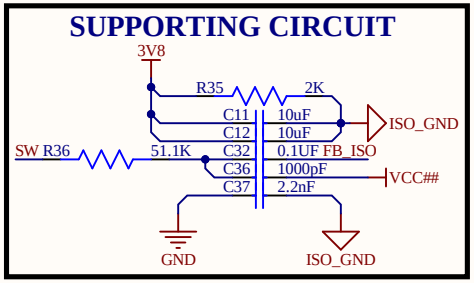
FEEDBACK NETWORK



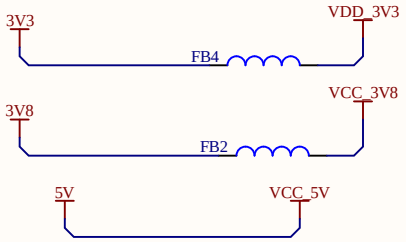
FLYBACK TRANSFORMER

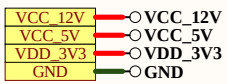


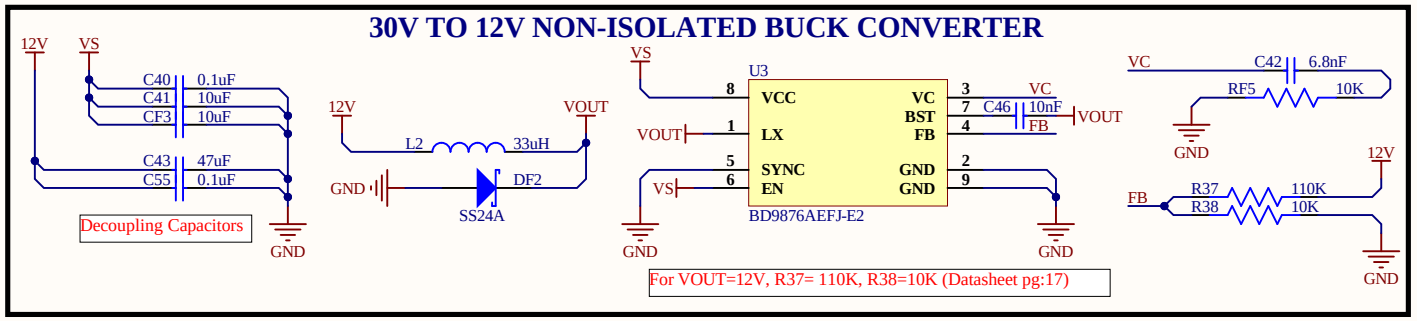
SUPPORTING CIRCUIT

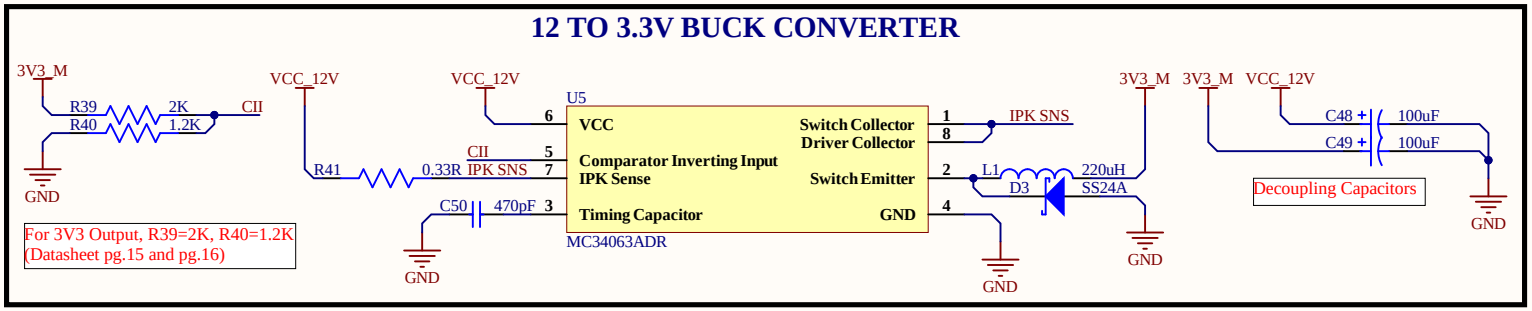


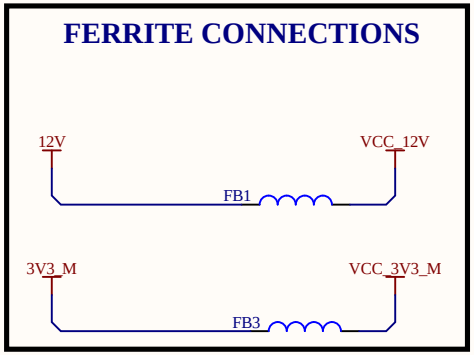
FERRITE CONNECTIONS

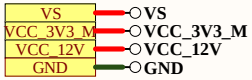


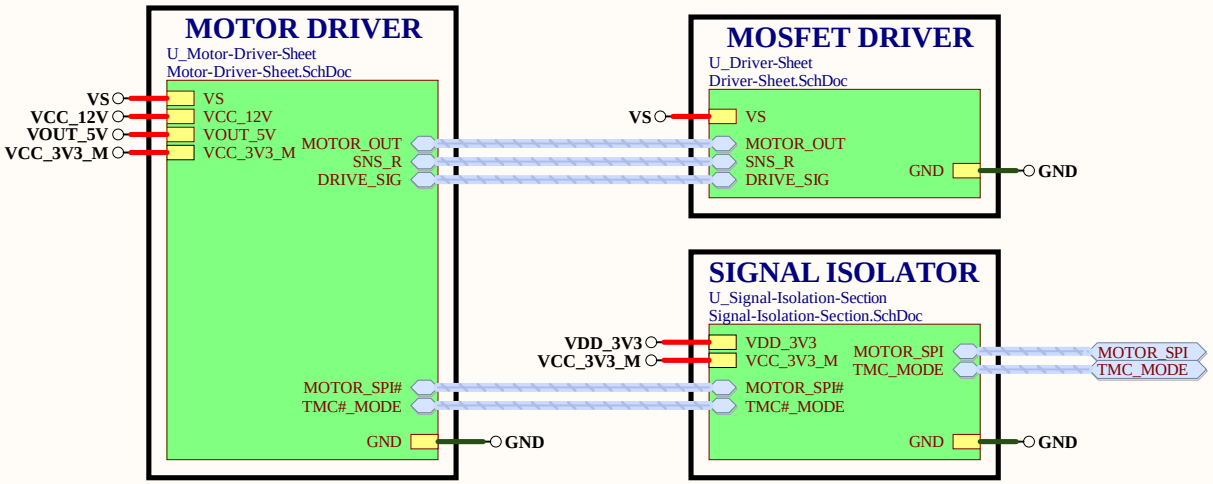


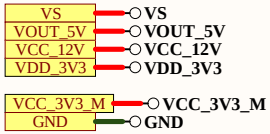

30V TO 12V NON-ISOLATED BUCK CONVERTER
12V
VS
C40 0.1uF
C41 10uF
CF3 10uF
C43 47uF
C55 0.1uF
Decoupling Capacitors
GND
12V
VOUT
L2 33uH
GND
DF2
SS24A
VS
U3
VCC
LX
SYNC
EN
VC
BST
FB
GND
GND
BD9876AEFJ-E2
VOUT
VC
C46 10nF
FB
VC
C42 6.8nF
RF5 10K
GND
12V
FB
R37 110K
R38 10K
GND
For VOUT=12V, R37= 110K, R38=10K (Datasheet pg:17)

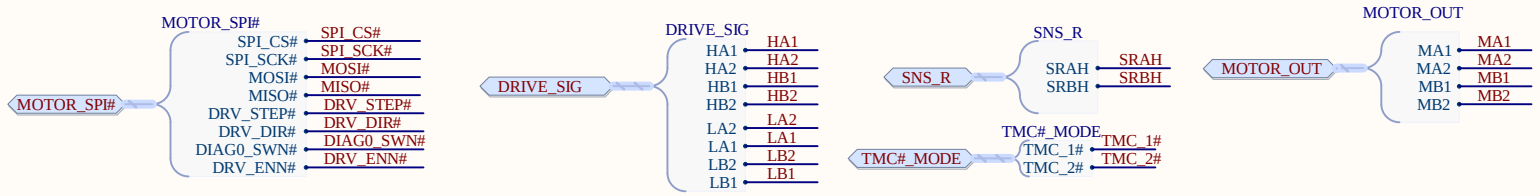

12 TO 3.3V BUCK CONVERTER
3V3_M
R39 2K
R40 1.2K
CII
GND
For 3V3 Output, R39=2K, R40=1.2K
(Datasheet pg.15 and pg.16)
VCC_12V
R41 0.33R
CII
IPK SNS
C50 470pF
GND
VCC_12V
U5
VCC
Comparator Inverting Input
IPK Sense
Timing Capacitor
Switch Collector
Driver Collector
Switch Emitter
GND
MC34063ADR
IPK SNS
L1 220uH
D3 SS24A
3V3_M
GND
GND
3V3_M
3V3_M
VCC_12V
C48 100uF
C49 100uF
Decoupling Capacitors
GND


FERRITE CONNECTIONS
12V
VCC_12V
FB1
3V3_M
VCC_3V3_M
FB3

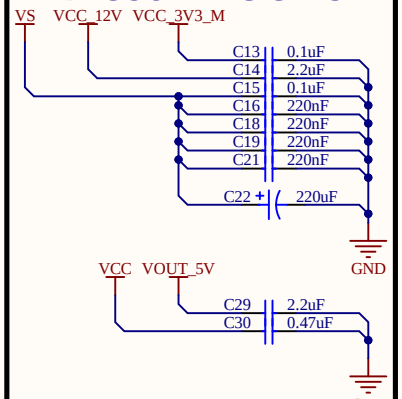

VS
VCC_3V3_M
VCC_12V
GND
VS
VCC_3V3_M
VCC_12V
GND


MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE


VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND



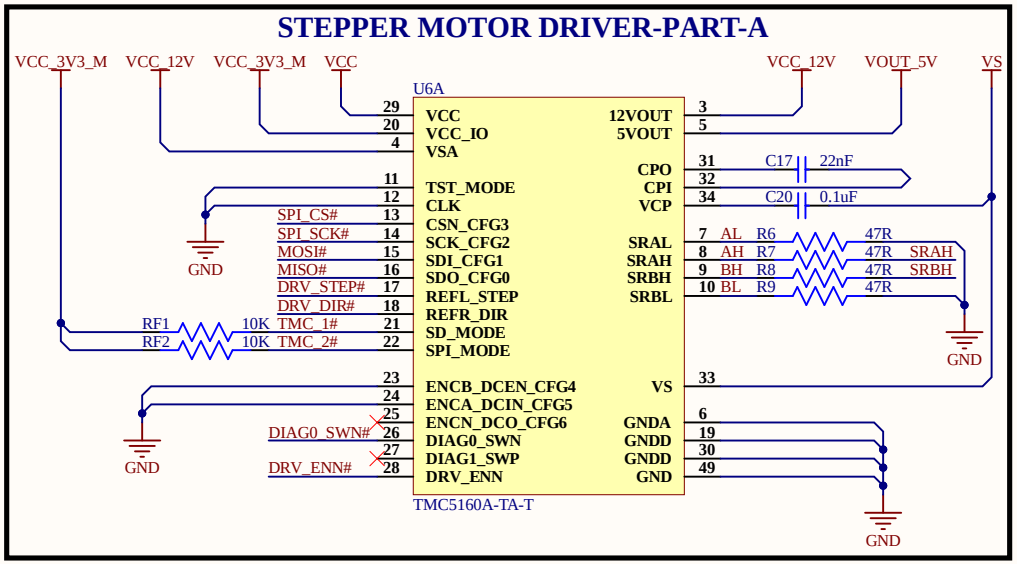
DECOUPLING CAPS



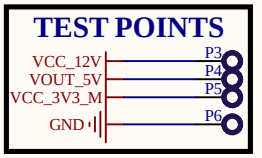
SERIES RESISTOR



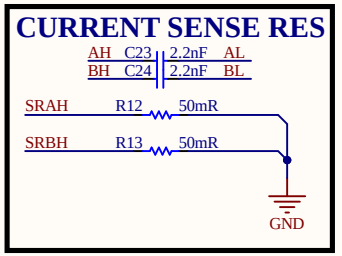
STEPPER MOTOR DRIVER-PART-A



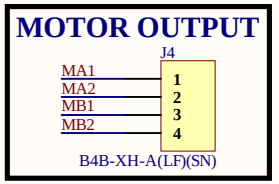
TEST POINTS



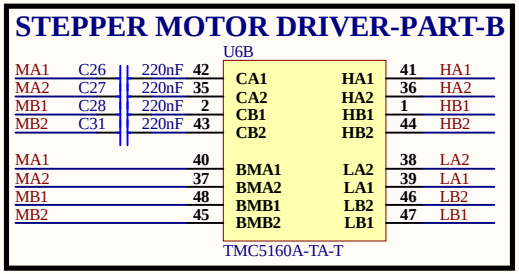
CURRENT SENSE RES



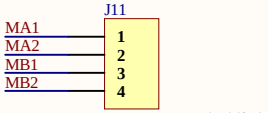
MOTOR OUTPUT

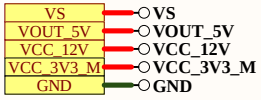


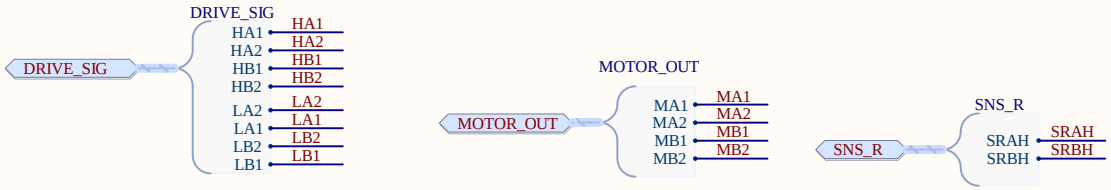
STEPPER MOTOR DRIVER-PART-B



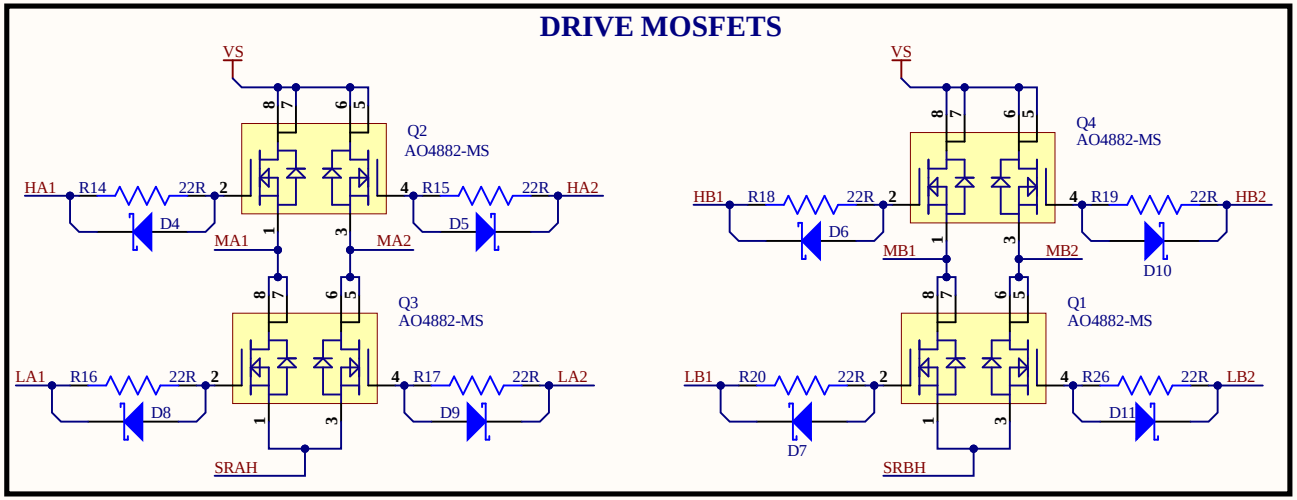
MOTOR OUTPUT FOR NEMA34



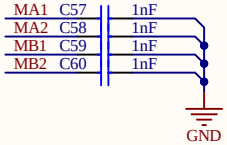




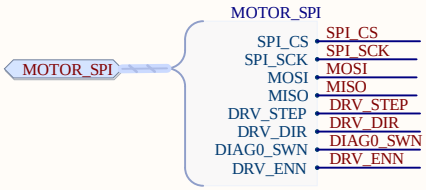
DRIVE MOSFETS

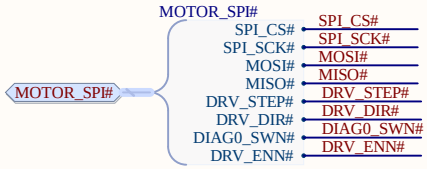


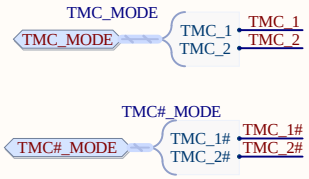
DECOUPLING CAPS

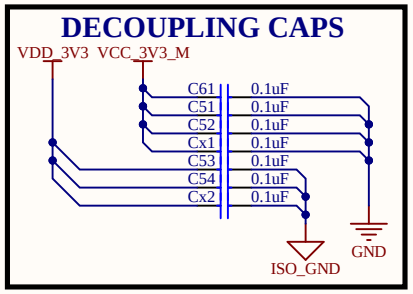


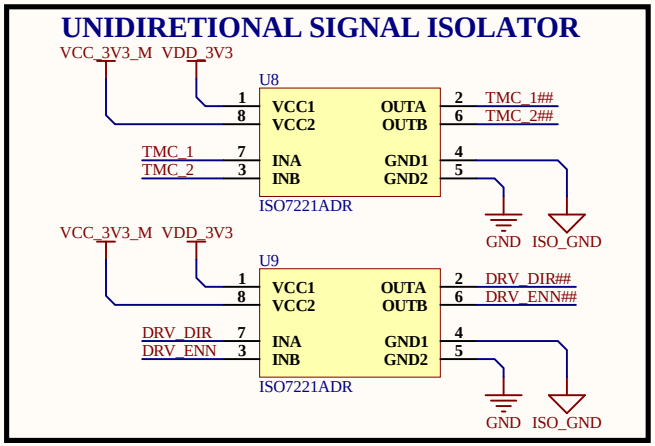


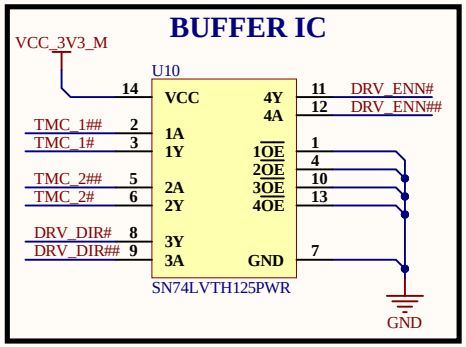


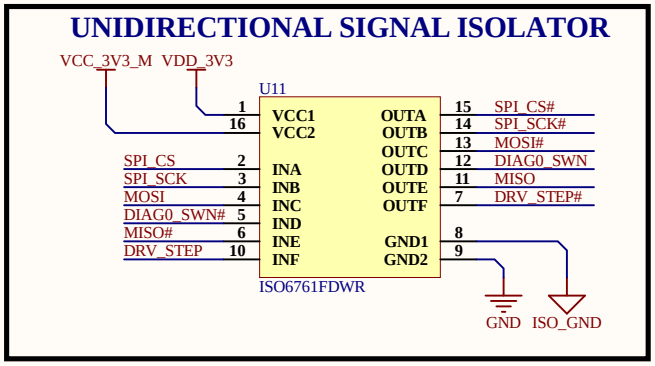


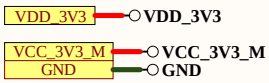








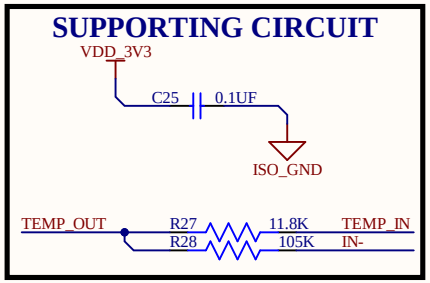




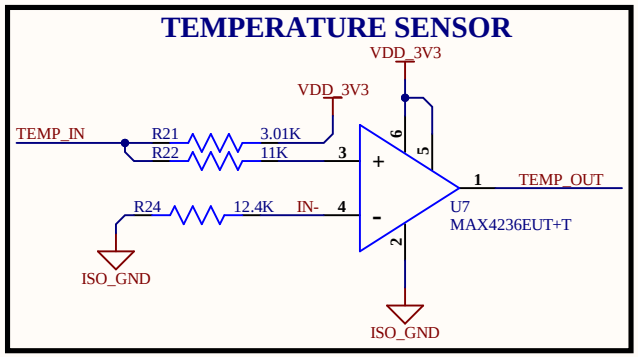
Title		
Size A4	Number	Revision
Date: 9/21/2023		Sheet of
File: C:\Users\..\Signal-Isolation-Section.SchDoc		Drawn By:

TEMP_OUT TEMP_OUT

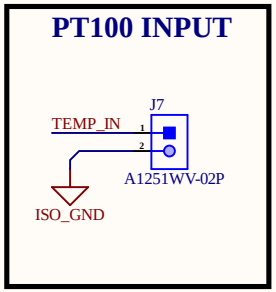
SUPPORTING CIRCUIT



TEMPERATURE SENSOR

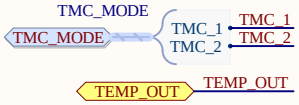


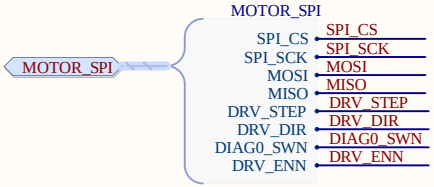
PT100 INPUT



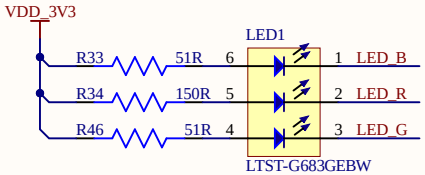
VDD_3V3 **VDD_3V3**
GND **GND**

INDICATION FOR MOTOR 5V SUPPLY

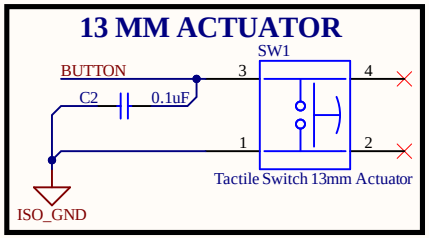




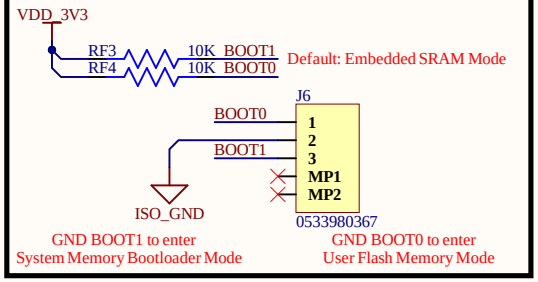
3.3V RGB LED



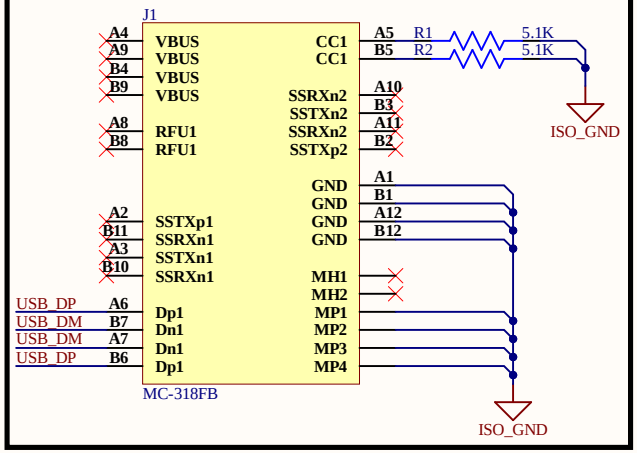
13 MM ACTUATOR



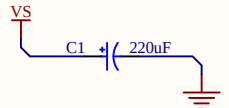
HEADERS TO SELECT BOOT MODE



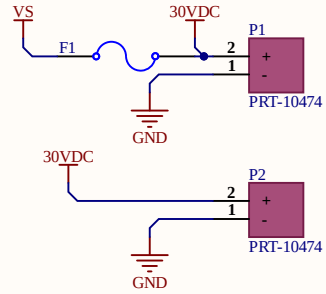
USB TYPE C FOR DATA TRANSFER



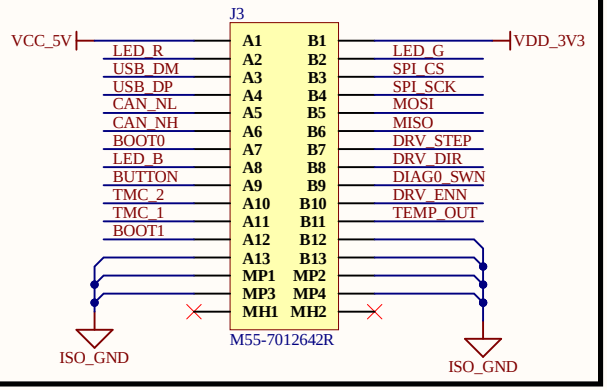
DECOUPLING CAP



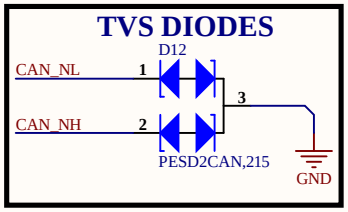
INPUT CONNECTORS



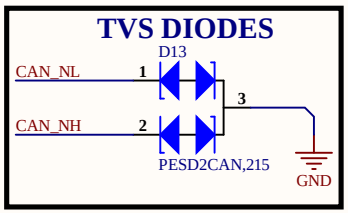
BOARD TO BOARD CONNECTOR



TVS DIODES



TVS DIODES



RJ45 FOR CAN BUS

